

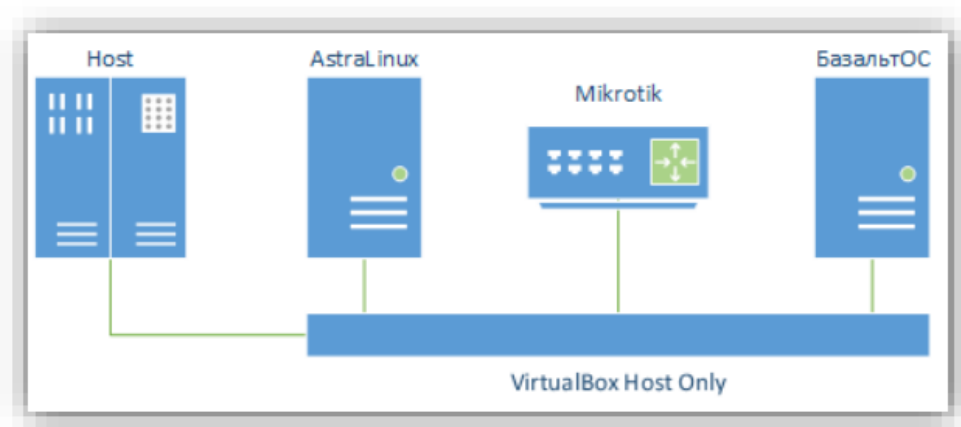
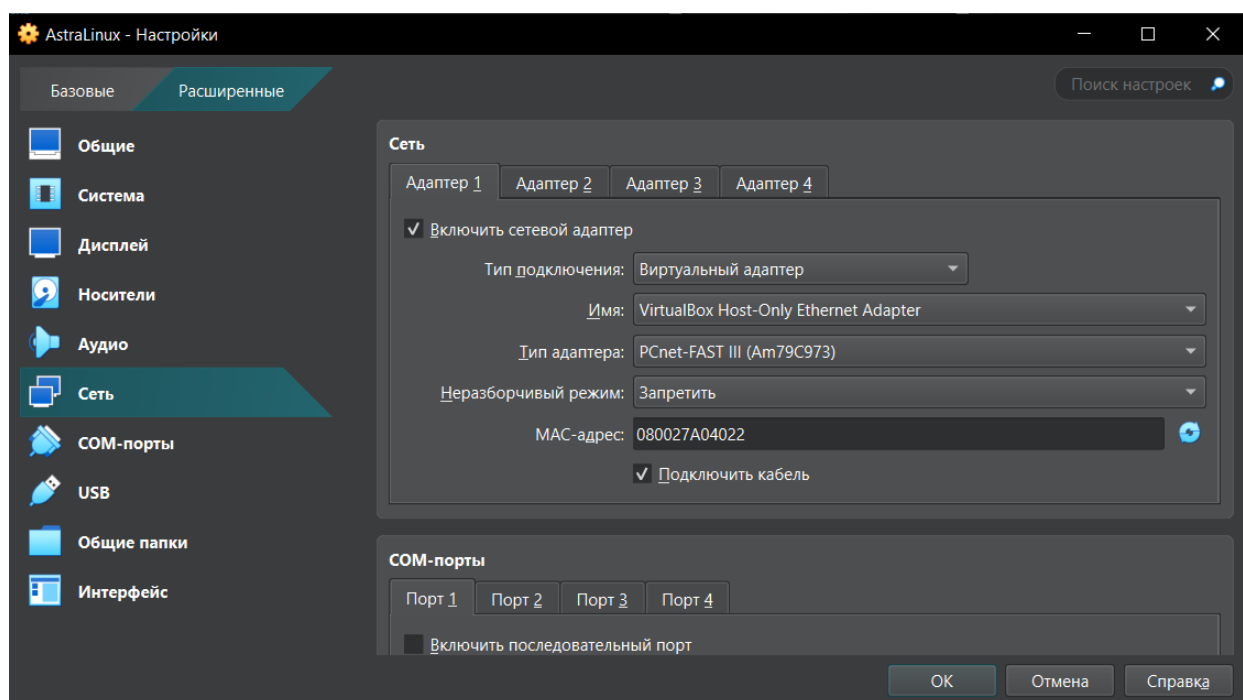
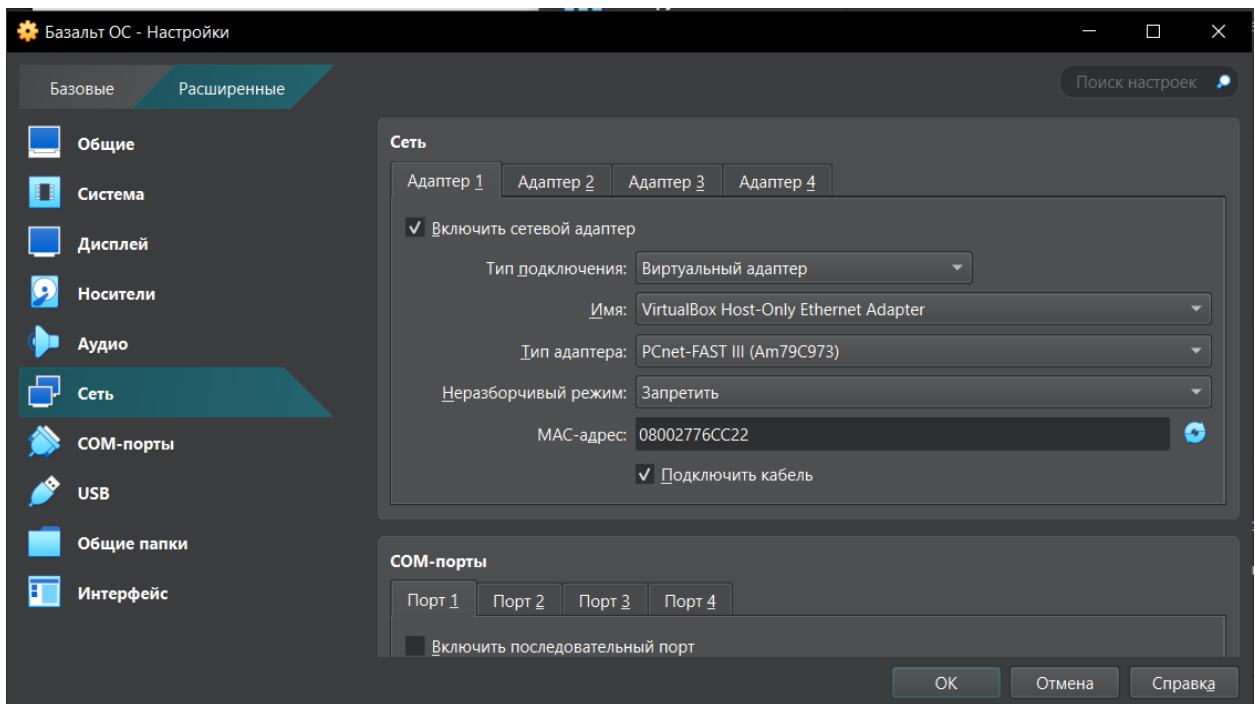
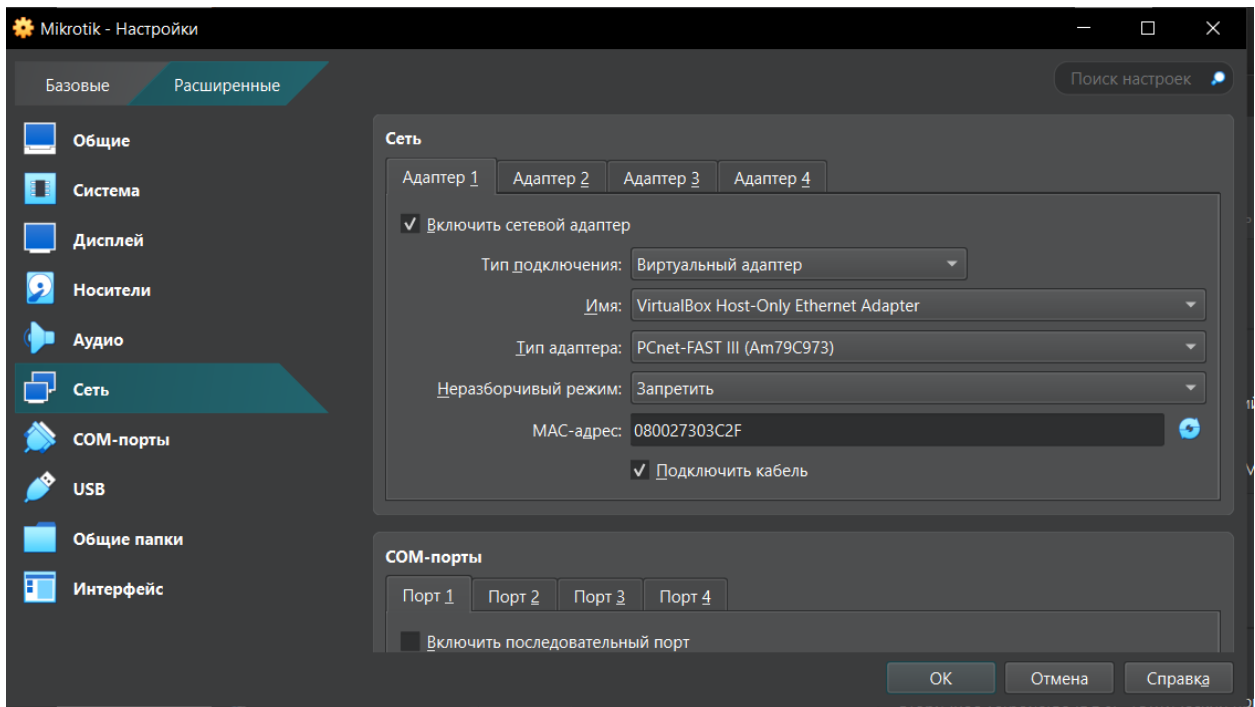


Рисунок 1 – Конфигурация сети для практического занятия

1.





2. link-local адрес (inet6) и MAC Базальта

```

[root@host-basalt ~]# ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
        valid_lft forever preferred_lft forever
[root@host-basalt ~]# ip a | grep "link/ether"
    link/ether 08:00:27:76:cc:22 brd ff:ff:ff:ff:ff:ff
[root@host-basalt ~]# ip -6 addr show | grep "inet6"
    inet6 ::1/128 scope host
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
[root@host-basalt ~]#

```

link-local адрес (inet6) и MAC AstraLinux

```

user@host-astra:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
user@host-astra:~$ ip -6 addr show | grep "inet6"
    inet6 ::1/128 scope host
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
user@host-astra:~$ ip -6 addr show | grep "link/ether"
user@host-astra:~$ ip a | grep "link/ether"
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
user@host-astra:~$

```

link-local адрес (inet6) и MAC MikroTik

```

[admin@MikroTik] > /ipv6 address print where address
Flags: D - DYNAMIC; L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
1 D ::1/128 lo no
2 DL fe80::a00:27ff:fe30:3c2f/64 ether1 no
[admin@MikroTik] > interface/print
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU MAC-ADDRESS
2 R ether1 ether 1500 08:00:27:30:3C:2F
1 R lo loopback 65536 00:00:00:00:00:00
[admin@MikroTik] > _

```

Хост (ipconfig /all) (link-local – Локальный адрес IPv6-адрес сети, MAC - физический адрес)

```

Адаптер Ethernet Ethernet:

DNS-суффикс подключения . . . . . :
Описание. . . . . : VirtualBox Host-Only Ethernet Adapter
Физический адрес. . . . . : 0A-00-27-00-00-05
DHCP включен. . . . . : Нет
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::627b:968f:bd5a:e8ba%5(Основной)
IPv4-адрес. . . . . : 10.16.65.2(Основной)
Маска подсети . . . . . : 255.255.192.0
Основной шлюз. . . . . :
IAID DHCPv6 . . . . . : 571080743
DUID клиента DHCPv6 . . . . . : 00-01-00-01-2B-83-A4-04-6C-24-08-D3-A2-1E
DNS-серверы. . . . . : fec0:0:0:ffff::1%1
                        fec0:0:0:ffff::2%1
                        fec0:0:0:ffff::3%1
NetBios через TCP/IP. . . . . : Включен

```

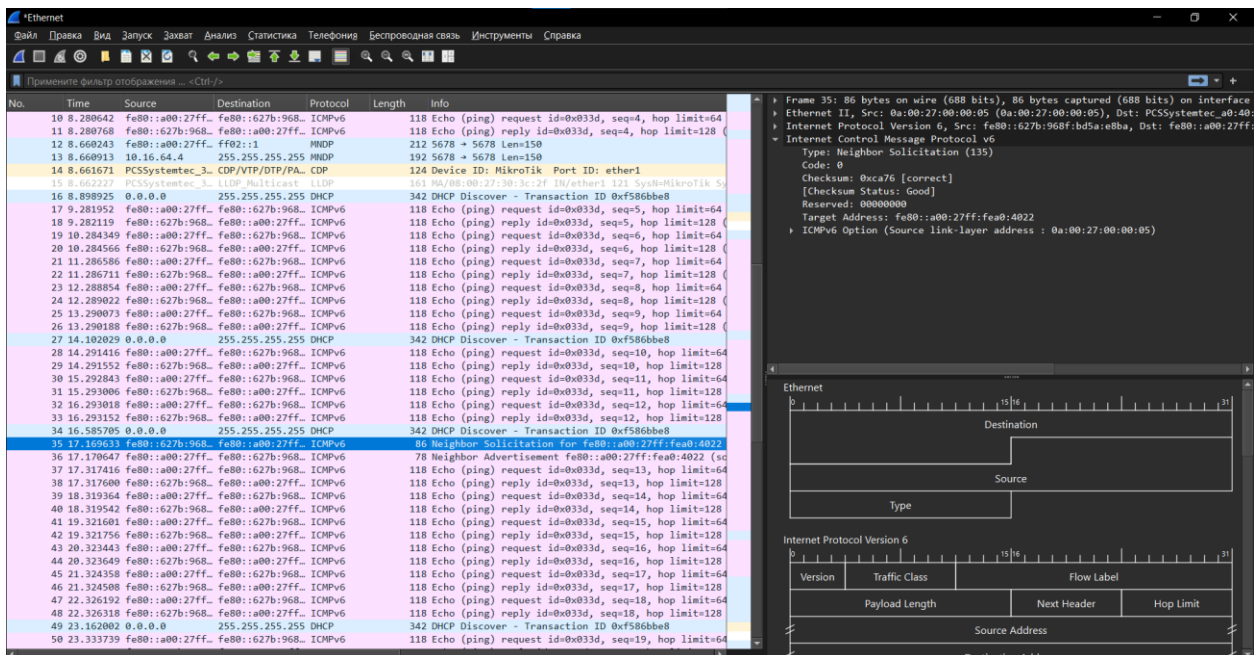
3. указываю eth0 после % так это интерфейс, через который нужно отправить пинг.

Пинг с Astra до хоста:

```

user@host-astra:~$ ping6 fe80::627b:968f:bd5a:e8ba%eth0
PING fe80::627b:968f:bd5a:e8ba%eth0(fe80::627b:968f:bd5a:e8ba%eth0) 56 data bytes
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=1 ttl=128 time=0.705 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=2 ttl=128 time=0.837 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=3 ttl=128 time=1.40 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=4 ttl=128 time=0.768 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=5 ttl=128 time=0.888 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=6 ttl=128 time=1.32 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=7 ttl=128 time=0.693 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=8 ttl=128 time=0.808 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=9 ttl=128 time=0.552 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=10 ttl=128 time=0.649 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=11 ttl=128 time=0.579 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=12 ttl=128 time=0.499 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=13 ttl=128 time=0.736 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=14 ttl=128 time=0.857 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=15 ttl=128 time=0.787 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=16 ttl=128 time=1.52 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=17 ttl=128 time=0.970 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=18 ttl=128 time=0.540 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=19 ttl=128 time=0.835 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=20 ttl=128 time=0.782 ms
^C
--- fe80::627b:968f:bd5a:e8ba%eth0 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19083ms
rtt min/avg/max/mdev = 0.499/0.837/1.528/0.274 ms
user@host-astra:~$

```



У протокола ICMPv6:

- **Neighbor Solicitation (NS)**— запрос на поиск MAC-адреса соседа
- **Neighbor Advertisement (NA)** — ответ с MAC-адресом
- **Echo Request и Echo Reply** — пакеты ping (отправки пинга на хост и ответ от хоста)

4.

```
user@host-astra:~$ ip -6 neigh show
fe80::627b:968f:bd5a:e8ba dev eth0 lladdr 0a:00:27:00:00:05 STALE
user@host-astra:~$
```

Команда **ip -6 neigh show** отображает таблицу соседей для IPv6. В этой таблице хранятся записи о соседних устройствах в сети (в этом случае сосед это хост, так как с ним было взаимодействие через пинг)

fe80::627b:968f:bd5a:e8ba :

Это **IPv6-адрес соседа**. link-local адрес хоста

dev eth0:

Указывает, через какой сетевой интерфейс (eth0) ваш узел взаимодействует с этим соседом.

lladdr 0a:00:27:00:00:05:

Это **MAC-адрес соседа**. Т.е. хоста можно убедиться выше
0a:00:27:00:00:05.

STALE:

Это состояние записи в таблице соседей. Состояние STALE означает, что запись о соседе есть, но она устарела (не использовалась некоторое время). Это нормальное состояние, если связь с соседом не активна в данный момент.

5. ping с маршрутизатора Mikrotik до узла AstraLinux (reply:
fe80::a00:27ff:fea0:4022):

```
[admin@MikroTik] > tool ping address=fe80::a00:27ff:fea0:4022%ether1
  SEQ HOST                                SIZE TTL TIME                        STATUS
  ---
0 fe80::a00:27ff:fea0:4022                56  64 806us                        echo reply
1 fe80::a00:27ff:fea0:4022                56  64 2ms917us                     echo reply
2 fe80::a00:27ff:fea0:4022                56  64 923us                       echo reply
3 fe80::a00:27ff:fea0:4022                56  64 1ms951us                     echo reply
4 fe80::a00:27ff:fea0:4022                56  64 2ms480us                     echo reply
5 fe80::a00:27ff:fea0:4022                56  64 2ms21us                      echo reply
6 fe80::a00:27ff:fea0:4022                56  64 1ms873us                     echo reply
7 fe80::a00:27ff:fea0:4022                56  64 2ms46us                      echo reply
8 fe80::a00:27ff:fea0:4022                56  64 2ms356us                     echo reply
9 fe80::a00:27ff:fea0:4022                56  64 2ms78us                      echo reply
10 fe80::a00:27ff:fea0:4022               56  64 2ms151us                     echo reply
11 fe80::a00:27ff:fea0:4022               56  64 2ms300us                     echo reply
12 fe80::a00:27ff:fea0:4022               56  64 2ms39us                      echo reply
13 fe80::a00:27ff:fea0:4022               56  64 2ms539us                     echo reply
14 fe80::a00:27ff:fea0:4022               56  64 3ms76us                      echo reply
15 fe80::a00:27ff:fea0:4022               56  64 2ms681us                     echo reply
sent=16 received=16 packet-loss=0% min-rtt=806us avg-rtt=2ms139us
max-rtt=3ms76us

[admin@MikroTik] >
```

ping с маршрутизатора Mikrotik до host-машины (request:
fe80::627b:968f:bd5a:e8ba):

```
[admin@MikroTik] > tool ping address=fe80::627b:968f:bd5a:e8ba%ether1
```

SEQ	HOST	SIZE	TTL	TIME	STATUS
0	fe80::627b:968f:bd5a:e8ba	56	128	5ms751us	echo reply
1	fe80::627b:968f:bd5a:e8ba	56	128	1ms17us	echo reply
2	fe80::627b:968f:bd5a:e8ba	56	128	918us	echo reply
3	fe80::627b:968f:bd5a:e8ba	56	128	695us	echo reply
4	fe80::627b:968f:bd5a:e8ba	56	128	566us	echo reply
5	fe80::627b:968f:bd5a:e8ba	56	128	653us	echo reply
6	fe80::627b:968f:bd5a:e8ba	56	128	1ms311us	echo reply
7	fe80::627b:968f:bd5a:e8ba	56	128	1ms850us	echo reply
8	fe80::627b:968f:bd5a:e8ba	56	128	793us	echo reply
9	fe80::627b:968f:bd5a:e8ba	56	128	1ms138us	echo reply
10	fe80::627b:968f:bd5a:e8ba	56	128	769us	echo reply
11	fe80::627b:968f:bd5a:e8ba	56	128	838us	echo reply
12	fe80::627b:968f:bd5a:e8ba	56	128	1ms181us	echo reply
13	fe80::627b:968f:bd5a:e8ba	56	128	1ms118us	echo reply
14	fe80::627b:968f:bd5a:e8ba	56	128	1ms88us	echo reply
15	fe80::627b:968f:bd5a:e8ba	56	128	1ms551us	echo reply
16	fe80::627b:968f:bd5a:e8ba	56	128	1ms527us	echo reply
17	fe80::627b:968f:bd5a:e8ba	56	128	970us	echo reply
18	fe80::627b:968f:bd5a:e8ba	56	128	395us	echo reply

```
sent=19 received=19 packet-loss=0% min-rtt=395us avg-rtt=1ms269us
max-rtt=5ms751us
```

Соседи MikroTik:

```
[admin@MikroTik] > ipv6 neighbor/print
```

```
Flags: R - router
```

0	address=fe80::a00:27ff:fe76:cc22	interface=ether1	mac-address=08:00:27:76:CC:22	status="stale"
1	address=fe80::a00:27ff:fea0:4022	interface=ether1	mac-address=08:00:27:A0:40:22	status="stale"
2	address=fe80::627b:968f:bd5a:e8ba	interface=ether1	mac-address=0A:00:27:00:00:05	status="stale"

```
[admin@MikroTik] > _
```

6. Повтор пунктов 3-5 для БазальтОС:

3. Пинг с БазальтОС до хоста:

```

[root@host-basalt ~]# ping6 fe80::627b:968f:bd5a:e8ba%enp0s3
PING fe80::627b:968f:bd5a:e8ba%enp0s3(fe80::627b:968f:bd5a:e8ba%enp0s3) 56 data bytes
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=1 ttl=128 time=3.50 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=2 ttl=128 time=1.35 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=3 ttl=128 time=0.783 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=4 ttl=128 time=0.876 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=5 ttl=128 time=1.07 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=6 ttl=128 time=1.13 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=7 ttl=128 time=0.965 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=8 ttl=128 time=1.06 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=9 ttl=128 time=0.870 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=10 ttl=128 time=1.09 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=11 ttl=128 time=2.35 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=12 ttl=128 time=1.64 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=13 ttl=128 time=1.09 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=14 ttl=128 time=0.933 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=15 ttl=128 time=1.02 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=16 ttl=128 time=1.19 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=17 ttl=128 time=0.945 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=18 ttl=128 time=0.978 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=19 ttl=128 time=1.15 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=20 ttl=128 time=1.59 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=21 ttl=128 time=0.959 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%enp0s3: icmp_seq=22 ttl=128 time=0.909 ms
^C
--- fe80::627b:968f:bd5a:e8ba%enp0s3 ping statistics ---
22 packets transmitted, 22 received, 0% packet loss, time 21026ms
rtt min/avg/max/meddev = 0.783/1.246/3.496/0.595 ms
[root@host-basalt ~]#

```

No.	Time	Source	Destination	Protocol	Length	Info
26	40.105889	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x99d5f85c
27	40.456526	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=4, hop limit=64 (reply in 28)
28	40.456641	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=4, hop limit=64 (request in 27)
29	41.457437	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=5, hop limit=64 (reply in 30)
30	41.457570	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=5, hop limit=64 (request in 29)
31	42.219341	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	86	Neighbor Solicitation for fe80::a00:27ff:fe76:cc22 from 0a:00:27:00:00:05
32	42.219585	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	78	Neighbor Advertisement for fe80::a00:27ff:fe76:cc22 (sol)
33	42.459840	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=6, hop limit=64 (reply in 34)
34	42.460023	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=6, hop limit=64 (request in 33)
35	43.461432	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=7, hop limit=64 (reply in 36)
36	43.461561	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=7, hop limit=64 (request in 35)
37	44.462311	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=8, hop limit=64 (reply in 38)
38	44.462467	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=8, hop limit=64 (request in 37)
39	45.463806	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=9, hop limit=64 (reply in 40)
40	45.463938	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=9, hop limit=64 (request in 39)
41	46.464748	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=10, hop limit=64 (reply in 42)
42	46.464916	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=10, hop limit=64 (request in 41)
43	46.909932	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x99d5f85c
44	47.466418	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=11, hop limit=64 (reply in 45)
45	47.466576	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=11, hop limit=64 (request in 44)
46	48.467095	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=12, hop limit=64 (reply in 47)
47	48.467928	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=12, hop limit=64 (request in 46)
48	49.468369	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=13, hop limit=64 (reply in 49)
49	49.468516	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=13, hop limit=64 (request in 48)
50	50.469394	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=14, hop limit=64 (reply in 51)
51	50.469523	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=14, hop limit=64 (request in 50)
52	51.470520	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=15, hop limit=64 (reply in 53)
53	51.470634	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=15, hop limit=64 (request in 52)
54	52.471391	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=16, hop limit=64 (reply in 55)
55	52.471551	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=16, hop limit=64 (request in 54)
56	53.472285	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=17, hop limit=64 (reply in 57)
57	53.472412	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=17, hop limit=64 (request in 56)
58	53.784260	fe80::a00:27ff:fe76:cc22	ff02::1	MNDP	212	5678 + 5678 Len=150
59	53.784879	10.16.64.4	255.255.255.255	MNDP	192	5678 + 5678 Len=150
60	53.785425	PCSSystemtec_3_	CDP/VTP/PAP/UDLD	CDP	124	Device ID: MikroTik Port ID: ether1
61	53.785901	PCSSystemtec_3_	LDP Multicast	LDP	161	0a/00:00:27:30:3c:2f 10/ether1 121 SysD=MikroTik SysD=MikroTik RouterOS 7.16.2 (stable) 202
62	53.857285	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x99d5f85c
63	54.473486	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=18, hop limit=64 (reply in 64)
64	54.473620	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=18, hop limit=64 (request in 63)
65	55.474405	fe80::a00:27ff:fe76:cc22	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x0001, seq=19, hop limit=64 (reply in 66)
66	55.474559	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fe76:cc22	ICMPv6	118	Echo (ping) reply id=0x0001, seq=19, hop limit=64 (request in 65)

4.

```

[root@host-basalt ~]# ip -6 neigh show
fe80::627b:968f:bd5a:e8ba dev enp0s3 lladdr 0a:00:27:00:00:05 STALE
[root@host-basalt ~]#

```

5. Пинг с Микротики до Базальта (reply: fe80::a00:27ff:fe76:cc22):


```
[admin@MikroTik] > ping fe80::a00:27ff:fe76:cc22%ether1
SEQ HOST                                SIZE TTL TIME                        STATUS
0 fe80::a00:27ff:fe76:cc22            56 64 3ms681us    echo reply
1 fe80::a00:27ff:fe76:cc22            56 64 1ms895us    echo reply
2 fe80::a00:27ff:fe76:cc22            56 64 2ms465us    echo reply
3 fe80::a00:27ff:fe76:cc22            56 64 2ms839us    echo reply
4 fe80::a00:27ff:fe76:cc22            56 64 2ms37us     echo reply
5 fe80::a00:27ff:fe76:cc22            56 64 2ms312us    echo reply
6 fe80::a00:27ff:fe76:cc22            56 64 2ms906us    echo reply
7 fe80::a00:27ff:fe76:cc22            56 64 2ms379us    echo reply
sent=8 received=8 packet-loss=0% min-rtt=1ms895us avg-rtt=2ms564us
max-rtt=3ms681us

[admin@MikroTik] > _
```

Пинг с Микротики до хоста:

```
[admin@MikroTik] > ping address=fe80::627b:968f:bd5a:e8ba%ether1
SEQ HOST                                SIZE TTL TIME                        STATUS
0 fe80::627b:968f:bd5a:e8ba            56 128 652us     echo reply
1 fe80::627b:968f:bd5a:e8ba            56 128 1ms216us    echo reply
2 fe80::627b:968f:bd5a:e8ba            56 128 1ms41us     echo reply
3 fe80::627b:968f:bd5a:e8ba            56 128 987us      echo reply
4 fe80::627b:968f:bd5a:e8ba            56 128 865us      echo reply
5 fe80::627b:968f:bd5a:e8ba            56 128 1ms189us    echo reply
6 fe80::627b:968f:bd5a:e8ba            56 128 1ms63us     echo reply
7 fe80::627b:968f:bd5a:e8ba            56 128 565us      echo reply
8 fe80::627b:968f:bd5a:e8ba            56 128 939us      echo reply
sent=9 received=9 packet-loss=0% min-rtt=565us avg-rtt=946us
max-rtt=1ms216us

[admin@MikroTik] > _
```

Соседи:

```
[admin@MikroTik] > ipv6 neighbor/print
Flags: R - router
0 address=fe80::a00:27ff:fe76:cc22 interface=ether1
  mac-address=08:00:27:76:CC:22 status="stale"

1 address=fe80::a00:27ff:fea0:4022 interface=ether1
  mac-address=08:00:27:A0:40:22 status="stale"

2 address=fe80::627b:968f:bd5a:e8ba interface=ether1
  mac-address=0A:00:27:00:00:05 status="stale"
[admin@MikroTik] >
```

7.

```
[admin@MikroTik] > ipv6 address/add address=fd00:2005:05:23::1/64 interface=ether1
[admin@MikroTik] > ipv6 nd print
Flags: X - disabled, I - invalid; * - default
0 * interface=all ra-interval=3m20s-10m ra-delay=3s mtu=unspecified
  reachable-time=unspecified retransmit-interval=unspecified
  ra-lifetime=30m ra-preference=medium hop-limit=unspecified
  advertise-mac-address=yes advertise-dns=yes
  managed-address-configuration=no other-configuration=no
[admin@MikroTik] > ipv6 address/print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 D ::1/128 lo no
1 DL fe80::a00:27ff:fe30:3c2f/64 ether1 no
2 G fd00:2005:5:23::1/64 ether1 yes
[admin@MikroTik] >
```

```
[admin@MikroTik] > ipv6 nd add interface=ether1
[admin@MikroTik] > ipv6 nd set [find interface=ether1] advertise-dns=yes advertise-mac=yes
[admin@MikroTik] > ipv6 nd print
Flags: X - disabled, I - invalid; * - default
0 * interface=all ra-interval=3m20s-10m ra-delay=3s mtu=unspecified
  reachable-time=unspecified retransmit-interval=unspecified
  ra-lifetime=30m ra-preference=medium hop-limit=unspecified
  advertise-mac-address=yes advertise-dns=yes
  managed-address-configuration=no other-configuration=no

1 interface=ether1 ra-interval=3m20s-10m ra-delay=3s mtu=unspecified
  reachable-time=unspecified retransmit-interval=unspecified
  ra-lifetime=30m ra-preference=medium hop-limit=unspecified
  advertise-mac-address=yes advertise-dns=yes
  managed-address-configuration=no other-configuration=no
[admin@MikroTik] > ipv6 address/print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 D ::1/128 lo no
1 DL fe80::a00:27ff:fe30:3c2f/64 ether1 no
2 G fd00:2005:5:23::1/64 ether1 yes
[admin@MikroTik] >
```

```

user@host-astra:~$ ping fd00:2005:05:23::1
PING fd00:2005:05:23::1(fd00:2005:5:23::1) 56 data bytes
64 bytes from fd00:2005:5:23::1: icmp_seq=1 ttl=64 time=2.48 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=2 ttl=64 time=2.19 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=3 ttl=64 time=1.85 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=4 ttl=64 time=2.22 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=5 ttl=64 time=1.82 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=6 ttl=64 time=2.32 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=7 ttl=64 time=1.60 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=8 ttl=64 time=1.75 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=9 ttl=64 time=1.41 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=10 ttl=64 time=2.29 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=11 ttl=64 time=2.02 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=12 ttl=64 time=1.76 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=13 ttl=64 time=2.55 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=14 ttl=64 time=2.19 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=15 ttl=64 time=3.45 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=16 ttl=64 time=2.46 ms
64 bytes from fd00:2005:5:23::1: icmp_seq=17 ttl=64 time=1.83 ms

```

(способ недействительный)

```

user@host-astra:~$ ping fd00:2005:05:23:a00:27ff:fea0:4022
PING fd00:2005:05:23:a00:27ff:fea0:4022(fd00:2005:5:23:a00:27ff:fea0:4022) 56 data bytes
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=1 ttl=64 time=0.067 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=4 ttl=64 time=0.055 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=5 ttl=64 time=0.059 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=6 ttl=64 time=0.061 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=7 ttl=64 time=0.070 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=8 ttl=64 time=0.045 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=9 ttl=64 time=0.063 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=10 ttl=64 time=0.059 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=11 ttl=64 time=0.082 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=12 ttl=64 time=0.061 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=13 ttl=64 time=0.124 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=14 ttl=64 time=0.058 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=15 ttl=64 time=0.091 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=16 ttl=64 time=0.055 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=17 ttl=64 time=0.064 ms
64 bytes from fd00:2005:5:23:a00:27ff:fea0:4022: icmp_seq=18 ttl=64 time=0.064 ms
^C
--- fd00:2005:05:23:a00:27ff:fea0:4022 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 17343ms
rtt min/avg/max/mdev = 0.045/0.065/0.124/0.020 ms
user@host-astra:~$ _

```

(способ действенный)

<Wireshark почему-то не выводит пакеты, хотя ошибок не возникает при пинге>

8.

```

user@host-astra:~$ sudo sysctl -w net.ipv6.conf.eth0.autoconf=1
net.ipv6.conf.eth0.autoconf = 1
user@host-astra:~$ ip -6 addr show dev eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591648sec preferred_lft 604448sec
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
user@host-astra:~$ ping6 fe80::627b:968f:bd5a:e8ba -I eth0
PING fe80::627b:968f:bd5a:e8ba(fe80::627b:968f:bd5a:e8ba) from fe80::a00:27ff:fea0:4022%eth0 eth0:
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=1 ttl=128 time=1.88 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=2 ttl=128 time=0.724 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=3 ttl=128 time=0.642 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=4 ttl=128 time=0.628 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=5 ttl=128 time=0.490 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=6 ttl=128 time=0.727 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=7 ttl=128 time=0.699 ms
^C
--- fe80::627b:968f:bd5a:e8ba ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6083ms
rtt min/avg/max/mdev = 0.490/0.827/1.881/0.437 ms
user@host-astra:~$

```

Завхват из Ethernet

Файл Правка Вид Запуск Завхват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения: <Ctrl>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e96d5d0
2	0.308563	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=1, hop limit=64 (reply in 3)
3	0.308663	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=1, hop limit=128 (request in 2)
4	1.426665	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=2, hop limit=64 (reply in 5)
5	1.426774	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=2, hop limit=128 (request in 4)
6	2.450418	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=3, hop limit=64 (reply in 7)
7	2.450561	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=3, hop limit=128 (request in 6)
8	2.793835	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e96d5d0
9	3.452643	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=4, hop limit=64 (reply in 10)
10	3.452773	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=4, hop limit=128 (request in 9)
11	4.454377	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=5, hop limit=64 (reply in 12)
12	4.454493	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=5, hop limit=128 (request in 11)
13	4.968006	fe80::a00:27ff:fea0:4022	ff02::1	MNDP	232	5678 + 5678 Lens=170
14	4.968379	10.16.64.4	255.255.255.255	MNDP	212	5678 + 5678 Lens=170
15	4.968851	PCSSystemtec_3...	VDP/DTIP/PAGP/UOLD	CDP	124	Device ID: MikroTik Port ID: ether1
16	4.969143	PCSSystemtec_3...	LDP/MLLS/LAS	LDP	187	0x00:00:27:00:00:0f 0x00:00:11:11:11:11 SysN-MikroTik SysD-MikroTik RouterOS 7.16.2 (stable) 2024-11-11
17	5.211386	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	86	Neighbor Solicitation for fe80::a00:27ff:fea0:4022 from 0a:00:27:00:00:05
18	5.212093	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	78	Neighbor Advertisement fe80::a00:27ff:fea0:4022 (sol)
19	5.458789	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=6, hop limit=64 (reply in 20)
20	5.458872	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=6, hop limit=128 (request in 19)
21	6.482971	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=7, hop limit=64 (reply in 22)
22	6.483089	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=7, hop limit=128 (request in 21)
23	7.485076	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=8, hop limit=64 (reply in 24)
24	7.485180	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=8, hop limit=128 (request in 23)
25	8.498267	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=9, hop limit=64 (reply in 26)
26	8.498375	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=9, hop limit=128 (request in 25)
27	8.919606	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e96d5d0
28	9.522363	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=10, hop limit=64 (reply in 29)
29	9.522451	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=10, hop limit=128 (request in 28)
30	10.546998	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=11, hop limit=64 (reply in 31)
31	10.547087	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=11, hop limit=128 (request in 30)
32	11.323476	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e96d5d0
33	11.570949	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=12, hop limit=64 (reply in 34)
34	11.571095	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=12, hop limit=128 (request in 33)
35	12.594079	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=13, hop limit=64 (reply in 36)
36	12.594183	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=13, hop limit=128 (request in 35)
37	13.618937	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=14, hop limit=64 (reply in 38)
38	13.619054	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=14, hop limit=128 (request in 37)
39	14.621409	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=15, hop limit=64 (reply in 40)
40	14.621557	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x03c9, seq=15, hop limit=128 (request in 39)
41	15.623338	fe80::a00:27ff:fea0:4022	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x03c9, seq=16, hop limit=64 (reply in 42)

Frame 1: 342 bytes on wire (Ethernet II, Src: PCSSystemtec_3... Internet Protocol Version 4, User Datagram Protocol, Src...

Internet Protocol Version 4

Version Header Len. Differentiator

Identification

Time to Live Prot

9.

```
user@host-astra:~$ sudo sysctl -w net.ipv6.conf.eth0.use_tempaddr=2
net.ipv6.conf.eth0.use_tempaddr = 2
user@host-astra:~$ ping6 fe80::627b:968f:bd5a:e8ba%eth0
PING fe80::627b:968f:bd5a:e8ba%eth0 (fe80::627b:968f:bd5a:e8ba%eth0) 56 data bytes
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=1 ttl=128 time=0.530 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=2 ttl=128 time=0.633 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=3 ttl=128 time=0.488 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=4 ttl=128 time=0.363 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=5 ttl=128 time=0.551 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=6 ttl=128 time=0.580 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=7 ttl=128 time=0.527 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=8 ttl=128 time=0.860 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=9 ttl=128 time=0.639 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=10 ttl=128 time=0.636 ms
^C
--- fe80::627b:968f:bd5a:e8ba%eth0 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9147ms
rtt min/avg/max/mdev = 0.363/0.580/0.860/0.125 ms
user@host-astra:~$ _
```

```
user@host-astra:~$ ip -6 route get fe80::627b:968f:bd5a:e8ba
fe80::627b:968f:bd5a:e8ba from :: dev eth0 proto kernel src fe80::a00:27ff:fea0:4022 metric 256 pre
f medium
user@host-astra:~$ _
```

src fe80::a00:27ff:fea0:4022—это IPv6 адрес источника.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xd01a5a5e
2	2.572900	fe80::a00:27ff::ff02::1	ff02::1	MNDP	232	5678 → 5678 Len=170
3	2.573280	10.16.64.4	255.255.255.255	MNDP	232	5678 → 5678 Len=170
4	2.573684	PCSSystemtec_3...	CDP/PTP/DP/PAgP/UDLD	CDP	124	Device ID: MikroTik Port ID: ether1
5	2.573956	PCSSystemtec_3...	LLDP/Multicast	LLDP	187	8A/00:00:27:30:3c:2f ID/ether1 121 System
6	5.366506	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04aa, seq=1, hop 1
7	5.366594	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x04aa, seq=1, hop 1
8	5.586260	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xd01a5a5e
9	6.368726	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04aa, seq=2, hop 1
10	6.368912	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x04aa, seq=2, hop 1
11	7.370701	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04aa, seq=3, hop 1
12	7.370794	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x04aa, seq=3, hop 1
13	7.509934	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xd01a5a5e
14	8.393129	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04aa, seq=4, hop 1
15	8.393228	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x04aa, seq=4, hop 1
16	9.417166	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04aa, seq=5, hop 1
17	9.417273	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x04aa, seq=5, hop 1
18	10.318056	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff:fea0:4022	ICMPv6	86	Neighbor Solicitation for fe80::a00:27ff:fea0:4022
19	10.318842	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::627b:968f:bd5a:e8ba	ICMPv6	78	Neighbor Advertisement fe80::a00:27ff:fea0:4022

Другой приоритет:

```
user@host-astra:~$ sudo sysctl -w net.ipv6.conf.eth0.use_tempaddr=1
net.ipv6.conf.eth0.use_tempaddr = 1
user@host-astra:~$ ping6 fe80::627b:968f:bd5a:e8ba%eth0
PING fe80::627b:968f:bd5a:e8ba%eth0 (fe80::627b:968f:bd5a:e8ba%eth0) 56 data bytes
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=1 ttl=128 time=0.442 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=2 ttl=128 time=0.396 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=3 ttl=128 time=0.485 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=4 ttl=128 time=0.480 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=5 ttl=128 time=0.429 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=6 ttl=128 time=0.526 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=7 ttl=128 time=1.13 ms
64 bytes from fe80::627b:968f:bd5a:e8ba%eth0: icmp_seq=8 ttl=128 time=0.810 ms
^C
--- fe80::627b:968f:bd5a:e8ba%eth0 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7147ms
rtt min/avg/max/mdev = 0.396/0.588/1.138/0.240 ms
user@host-astra:~$ _
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8113953a
2	2.239570	fe80::a00:27ff::ff02::1	ff02::1	MNDP	232	5678 → 5678 Len:170
3	2.240024	10.16.64.4	255.255.255.255	MNDP	212	5678 → 5678 Len:170
4	2.240396	PCSSystemtec_3...	CDP/VTP/DTP/PagP/UDLD	CDP	124	Device ID: MikroTik Port ID: ether1
5	2.240726	PCSSystemtec_3...	CDP/VTP/DTP/PagP/UDLD	CDP	124	Device ID: MikroTik Port ID: ether1
6	2.667851	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=1, hop limit=64 (r
7	2.667942	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=1, hop limit=128 (r
8	3.083982	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8113953a
9	3.673116	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=2, hop limit=64 (r
10	3.673243	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=2, hop limit=128 (r
11	4.674806	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=3, hop limit=64 (r
12	4.674939	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=3, hop limit=128 (r
13	5.494879	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8113953a
14	5.676422	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=4, hop limit=64 (r
15	5.676548	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=4, hop limit=128 (r
16	6.680074	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=5, hop limit=64 (r
17	6.680178	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=5, hop limit=128 (r
18	7.704185	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=6, hop limit=64 (r
19	7.704298	fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) reply id=0x04af, seq=6, hop limit=128 (r
20	8.728822	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	fe80::a00:27ff::fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x04af, seq=7, hop limit=64 (r

```

Frame 9: 118 bytes on wire (944 bits), 118 bytes captured (944 bi
Ethernet II, Src: PCSSystemtec_a0:40:22 (08:00:27:a0:40:22), Dst:
Internet Protocol Version 6, Src: fe80::a00:27ff::fe80:4022, Dst:
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00
.... 1110 0100 0100 1110 = Flow Label: 0xb44e
Payload Length: 64
Next Header: ICMPv6 (58)
Hop Limit: 64
Source Address: fe80::a00:27ff::fe80:4022
Destination Address: fe80::627b:968f:bd5a:e8ba
[Source SLAAC MAC: PCSSystemtec_a0:40:22 (08:00:27:a0:40:22)]
[Stream index: 1]
Internet Control Message Protocol v6
Type: Echo (ping) request (128)
Code: 0
Checksum: 0xc9aa [correct]
[Checksum Status: Good]
Identifier: 0x04af
Sequence: 2

```

10. Удаление назначенный IPv6 адрес на Mikrotik:

```

[admin@MikroTik] > ipv6 address/print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 G fd00:2005:5:23::1/64 ether1 yes
1 D ::1/128 lo no
2 DL fe80::a00:27ff::fe30:3c2f/64 ether1 no
[admin@MikroTik] > ipv6 address/remove number=0
[admin@MikroTik] > ipv6 address/print
Flags: D - DYNAMIC; L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 D ::1/128 lo no
1 DL fe80::a00:27ff::fe30:3c2f/64 ether1 no
[admin@MikroTik] > _

```

Настройка Астры:

```

user@host-astra:~$ sudo ip -6 addr add fd00:2005:05:23::2/64 dev eth0
user@host-astra:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group default qlen 1000
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
    inet 169.254.4.217/16 brd 169.254.255.255 scope link eth0:avahi
        valid_lft forever preferred_lft forever
    inet6 fd00:2005:5:23::2/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fd00:2005:5:23:7fe:3c06:76d3:d006/64 scope global temporary deprecated dynamic
        valid_lft 604334sec preferred_lft 0sec
    inet6 fd00:2005:5:23:a00:27ff::fe80:4022/64 scope global deprecated mngtppaddr dynamic
        valid_lft 2591867sec preferred_lft 0sec
    inet6 fe80::a00:27ff::fe80:4022/64 scope link
        valid_lft forever preferred_lft forever
user@host-astra:~$

```


Настройка полученного интерфейса на получение IPV6 адреса по dhcpv6:

```
[admin@MikroTik] > ipv6 dhcp-client add interface=ether1 request=prefix pool-name=dhcp-pool
[admin@MikroTik] > _
```

Пинг до Астры:

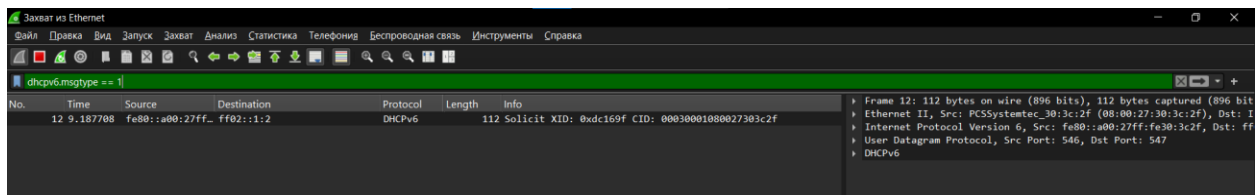
```
user@host-astra:~$ sudo ip -6 addr add fd:2005:05:23::2/64 dev eth0
user@host-astra:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group default qlen 1000
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
    inet 169.254.4.217/16 brd 169.254.255.255 scope link eth0:avahi
        valid_lft forever preferred_lft forever
    inet6 fd:2005:5:23::2/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fd:2005:5:23:230f:a139:aaa8:5e86/64 scope global temporary dynamic
        valid_lft 604632sec preferred_lft 85965sec
    inet6 fd:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591833sec preferred_lft 604633sec
    inet6 fd00:2005:5:23:7fe:3c06:76d3:d006/64 scope global temporary dynamic
        valid_lft 601636sec preferred_lft 82969sec
    inet6 fd00:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591833sec preferred_lft 604633sec
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
user@host-astra:~$ _
```

```
[admin@MikroTik] > ping fd:2005:5:23::2
  SEQ HOST                SIZE TTL TIME          STATUS
    0 fd:2005:5:23::2      56  64 1ms767us    echo reply
    1 fd:2005:5:23::2      56  64 1ms34us     echo reply
    2 fd:2005:5:23::2      56  64 664us       echo reply
    3 fd:2005:5:23::2      56  64 854us       echo reply
    4 fd:2005:5:23::2      56  64 989us       echo reply
    5 fd:2005:5:23::2      56  64 1ms192us    echo reply
sent=6 received=6 packet-loss=0% min-rtt=664us avg-rtt=1ms83us
max-rtt=1ms767us
[admin@MikroTik] >
```

Освобождение и запрос адреса на Микротик:

```
[admin@MikroTik] > ipv6 dhcp-client/release [find interface=ether1]
[admin@MikroTik] > ipv6 dhcp-client/renew [find interface=ether1]
[admin@MikroTik] > _
```

Solicit: Клиент отправляет запрос на поиск DHCPv6 сервера. Фильтр в Wireshark: dhcpv6.msgtype == 1



Захват из Ethernet						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
157	77.691589	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=51, hop limit=128 (requ
158	78.718375	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=52, hop limit=64 (rep
159	78.718464	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=52, hop limit=128 (requ
160	79.741110	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=53, hop limit=64 (rep
161	79.741193	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=53, hop limit=128 (requ
162	80.766409	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=54, hop limit=64 (rep
163	80.766505	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=54, hop limit=128 (requ
164	81.764635	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xef638b41
165	81.786436	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=55, hop limit=64 (rep
166	81.786519	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=55, hop limit=128 (requ
167	82.815197	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=56, hop limit=64 (rep
168	82.815316	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=56, hop limit=128 (requ
169	83.838946	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=57, hop limit=64 (rep
170	83.839047	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=57, hop limit=128 (requ
171	84.437724	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	86	Neighbor Solicitation for fe80::a00:27ff:fea0:4022 from
172	84.438032	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	78	Neighbor Advertisement fe80::a00:27ff:fea0:4022 (sol)
173	84.859125	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=58, hop limit=64 (rep
174	84.859228	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=58, hop limit=128 (requ
175	85.887129	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=59, hop limit=64 (rep
176	85.887218	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=59, hop limit=128 (requ
177	86.249466	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xef638b41
178	86.908257	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=60, hop limit=64 (rep
179	86.908353	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=60, hop limit=128 (requ
180	87.935569	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=61, hop limit=64 (rep
181	87.935812	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=61, hop limit=128 (requ
182	88.449975	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xef638b41
183	88.939462	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=62, hop limit=64 (rep
184	88.939565	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=62, hop limit=128 (requ
185	89.951367	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=63, hop limit=64 (rep
186	89.951470	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=63, hop limit=128 (requ
187	90.972960	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=64, hop limit=64 (rep
188	90.973039	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=64, hop limit=128 (requ
189	91.998837	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=65, hop limit=64 (rep
190	91.998926	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=65, hop limit=128 (requ
191	93.022825	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=66, hop limit=64 (rep
192	93.022936	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=66, hop limit=128 (requ
193	94.046818	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=67, hop limit=64 (rep
194	94.046905	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=67, hop limit=128 (requ
195	95.070857	fe80::a00:27ff...	fe80::627b:968f:bd5a:e8ba	ICMPv6	118	Echo (ping) request id=0x051f, seq=68, hop limit=64 (rep
196	95.070945	fe80::627b:968...	fe80::a00:27ff:fea0:4022	ICMPv6	118	Echo (ping) reply id=0x051f, seq=68, hop limit=128 (requ
197	95.224162	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xef638b41